GRIP STRENGTH AND CARDIOVASCULAR HEALTH

RESEARCH QUESTION

Is hand grip strength associated with cardiovascular markers independent of age, gender, and BMI?

HYPOTHESIS (from literature)

Low grip strength (sarcopenia/frailty) → worse cardiovascular health

STUDY DESIGN

- Cross-sectional analysis of HPP 10K cohort
- Age range: 40-70 years
- Complete case analysis

EXPOSURE

- Hand grip strength (average of left and right hand)
- Measured by dynamometer

OUTCOMES

- Pulse Wave Velocity (PWV) - arterial stiffness
- Blood Pressure (systolic, diastolic)
- Intima-Media Thickness (IMT) - carotid atherosclerosis
- Ankle-Brachial Index (ABI) - peripheral artery disease

CONFOUNDERS CONTROLLED

- Age (continuous)
- Gender (binary)
- BMI (continuous)

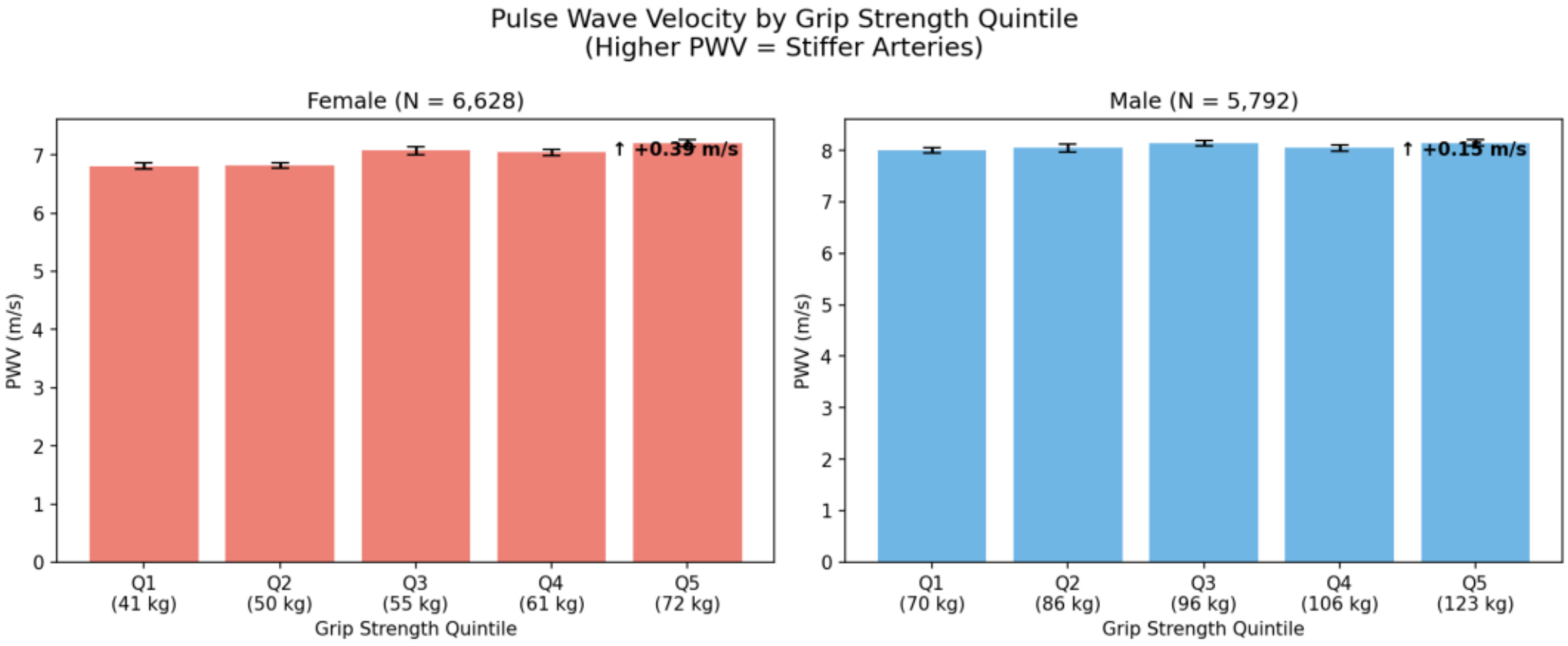
ANALYSIS

- Partial correlations (controlling for confounders)
- Multiple linear regression
- Gender-stratified analysis
- Age-stratified analysis

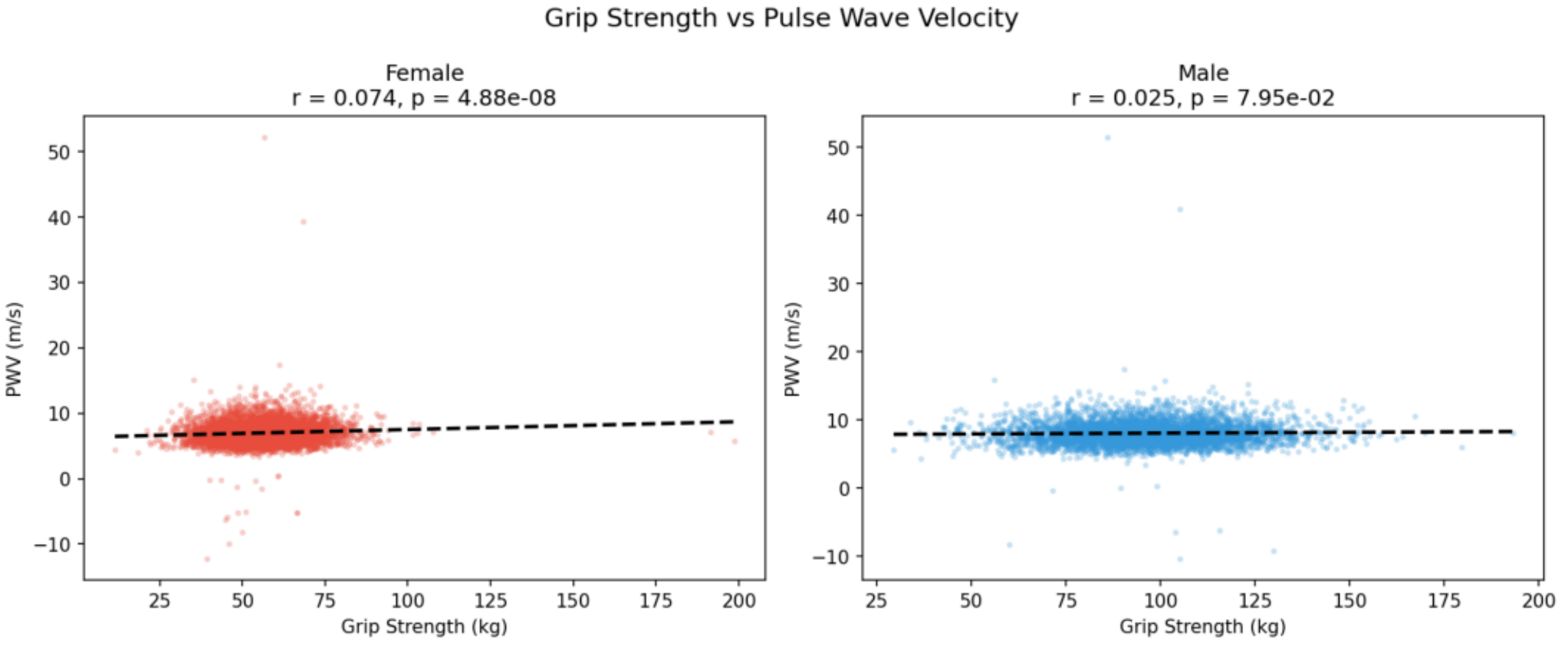
KEY FINDING: UNEXPECTED

Higher grip strength is associated with HIGHER PWV and blood pressure, OPPOSITE to the frailty hypothesis from literature!

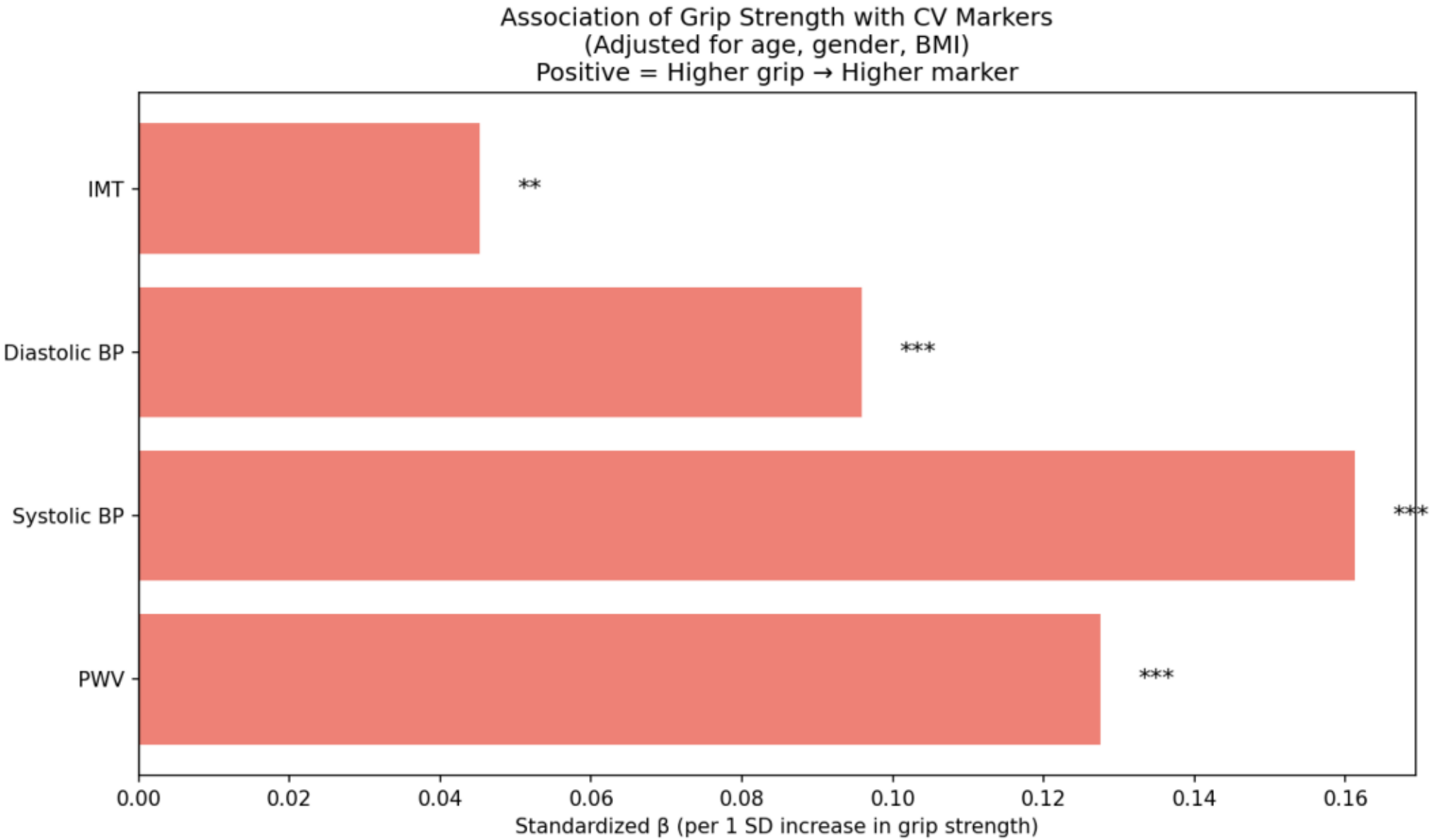
Main Finding: Higher Grip → Higher PWV



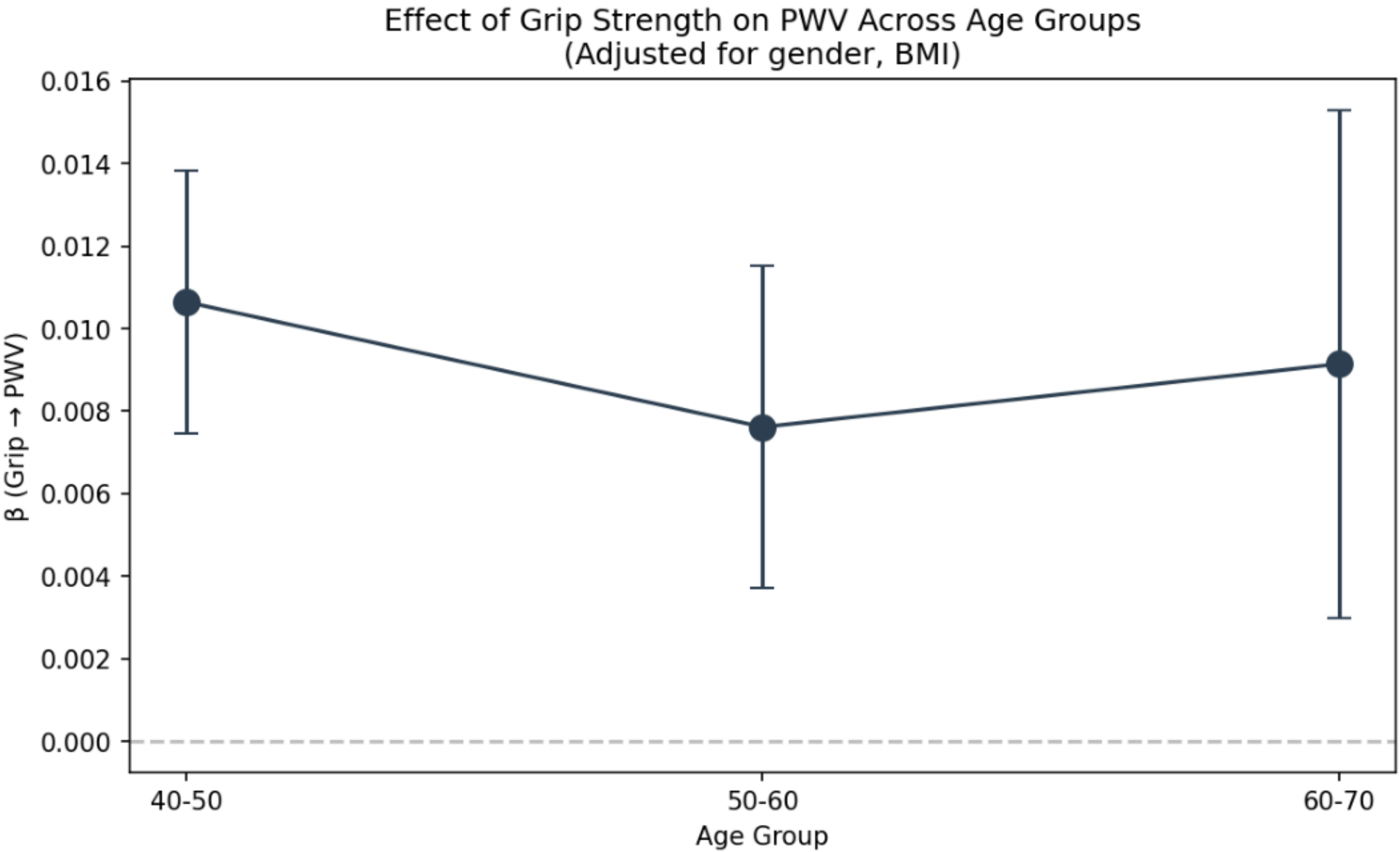
Scatter: Grip Strength vs PWV



Effect Sizes Across CV Markers



Age-Stratified Effects



KEY FINDINGS SUMMARY

SAMPLE: N = 12,371 (ages 40-70, 47% male)

MAIN RESULT: OPPOSITE TO HYPOTHESIS
Higher grip strength is associated with HIGHER (worse) cardiovascular markers, not lower as predicted by the frailty literature.

ADJUSTED ASSOCIATIONS (controlling for age, gender, BMI)

Outcome	Partial r	β	p-value
Pulse Wave Velocity	+0.081	+0.0096	< 0.001 ***
Systolic BP	+0.111	+0.1075	< 0.001 ***
Diastolic BP	+0.061	+0.0418	< 0.001 ***
Intima-Media Thickness	+0.030	+0.0002	0.002 **
Ankle-Brachial Index	+0.017	+0.0001	0.056 (NS)

INTERPRETATION
- In this healthy population, grip strength is a marker of physical activity and muscle mass, associated with higher cardiac output and sympathetic tone → higher blood pressure and PWV

- The frailty-CV mortality association from longitudinal studies may not translate to cross-sectional CV marker associations in healthy middle-aged populations

- Effect is consistent across:
  - Both genders (stronger in females)
  - All age groups (40-50, 50-60, 60-70)

CLINICAL IMPLICATIONS
- Do NOT interpret low grip strength as "better" CV health
- The grip strength-CV marker relationship is complex and may differ from the grip strength-CV mortality relationship
- More research needed on mechanisms

LIMITATIONS
1. Cross-sectional design - cannot establish causality
2. Healthy cohort - may not generalize to frail populations
3. Grip strength units unclear (may be summed trials)
4. Did not control for physical activity, medications